

Power Consumption Monitoring in ITIM Datacenter

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Abstract— This paper describe the temperature monitoring and the possibilities to reduce the power consumption in Datacenter. The National Institute for Research and Development of Isotopic and Molecular Technology (ITIM) datacenter from Cluj-Napoca has some special function. It incorporates a Grid site, and High performance site and a simple network. The monitoring function is designed to monitor the power consumption off this 3 system.

Keywords - monitoring, power consumption, computer system

I. INTRODUCTION

In recent years, computer systems have been increasingly applied in industry and research. These were used both for monitoring various operations, and simulation of processes and technologies. With the development of their knowledge of the environmental conditions was necessary for operation with minimal energy consumption and failures preventing. To this end, studies have shown that an important role in the smooth functioning of computer systems is the ambient temperature, humidity, wetness, airflow etc [1-3].

The air conditioning has a very important role in keeping these parameters in optimal operating limits of the systems studied. Because the air conditioning plays a central role in the successful operation of the computer systems, this requires careful monitoring. When the air conditioning can't be directly monitored, the best proxy for an air conditioner failing is the cold air inlet from the air conditioner into the server room cold aisle. Any failure by the air conditioner will be most noticeable here because the difference in temperature between the hot isle and the cold isle is greatest at the cold air inlet. Consequently, you can use a lower temperature threshold for your alarm than you might otherwise be able to elsewhere [4].

To maintain the temperature in computer systems and datacenters, around 20 Celsius degree with maximum deviation of 4 Celsius degree, it needs to monitor the temperature in the server room all the time, especially at night and weekends when nobody is around [5-7]. It need to measure temperatures at different points in the room to get an idea of where the hot spots might be. It needs temperature measurement to be automated and reliable, so a network attached device is ideal. The device must support alarms, via a number of different methods like email or SMS. It should be able to set the system up and then be alerted when something is wrong [6].

Until very recently, power consumption, management and monitoring have neither required nor received any attention. The great consumption of power in the last years, the high

prices paid for electric power and the impact in environment are the basic reason why people and business around the world are trying to monitor and manage in better ways their own power consumption. Solutions exist in all domains especially for big enterprises. But what happen for medium enterprises?

A simple answer to this question would be, we are trying to acquire our solution or build them our self. For what do we need such solution?

At the National Institute for Research and Development of Isotopic and Molecular Technology from Cluj-Napoca is functioning a datacenter which. The datacenter is occupied by a maximum of 10 racks which are processing a lot of jobs from around the world.

The Institute datacenter has a Grid site named RO-14-ITIM from which 95% is dedicated to experiment ATLAS at the European Organization for Nuclear Research, known as CERN, a High Performance Computing cluster which is processing physics jobs and a network which is applied for the Internet functionality of the Institute. The High Performance Computing cluster is processing jobs in many research domains which are part of the institute research area: the molecule-based and molecule-controlled electronic device research, research in computational simulation for biological molecules and the processing of solid NMR Parameters using Quantum ESPRESSO [8]. Each of this domain represent another research area of the Institute.

The datacenter has a capacity of 10 racks; each of them is almost full of computers. Today we have 4 cooling air devices in the datacenter, over 1000 core which has to be cooled and a network consisting of a 6509E Cisco switch. The storage capacity of the datacenter is 200 TB dedicated to the High performance cluster and the Grid site.

For the last four years problem occurs in the datacenter and in the surrounding laboratories because of the great power consumption. Many questions were raised but no answer were found. In the last year a team made of 5 people got the problem in their hands and a special experimental device is being made for monitoring and managing the power consumption. In parallel with the cluster we are monitoring the temperature in the datacenter and the humidity.

In this paper we monitored the operation of air conditioning equipments, the temperature variations and power consumption in INCDTIM Datacenter, in order to establish the methods for reducing power consumption.

In our previously work we performed a monitoring system that consisted in periodic review of those parameters that are not watched by the system default, but whose evolution over time can provide information on that and can predict the occurrence of events [9]. The system was focused on identifying undesirable events as soon as possible, which usually causes the system out of operation and which are typically caused by partial or total failure of sensors or actuators. Unit automation also involves the regulation of the system power consumption.

II. MONITORING SYSTEM

This application consists of watching a user selected frequency in the range [0.1 ... 60] second of the three monitored parameters for each component on control system. The following parameters are:

1. voltage power supply of the cooling units;
2. state of rotation of the main outside fans that serves the unit, with their propeller motion detection;
3. temperature of the heat transfer of heat exchange circuit.

Tracking is done through a computer program that allows viewing parameters previously described into a suggestive and graphic interface which also allows easy control of an alarm device which consists of four electronic lights and a buzzer (Buzzer). Parameter values are provided by specialized sensors mounted on the system monitored by a signal taken by boards and processed by a computer program. It displays the values determined through a dedicated interface and triggers the ALARM based on these values. Program execution is created and launched "virtual device with security". Below is the construction of the device both in terms of "hardware" and "software", explains the operation of its [9].

The hardware of equipment consists of a development board, equipped with eight digital outputs and four digital inputs and four analog inputs, with its "firmware" called "AVR_NET_IO" (Fig. 1) that using a microcontroller ATmega32 type. adaptation electronic circuit..

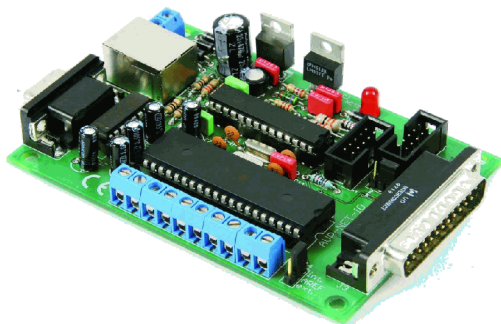


Fig. 1 AVR_NET_IO electronic board

This was completed with an adaptation circuit (a digital multiplexer) needed to connect eight sensors to the four-channel digital acquisition circuit (four sensors for operating state of cooling units and four sensors to monitor the main fan rotation). Temperatures are taken directly to the development board using four analog inputs where measurement is 10 bit

resolution analog signal provided by integrated temperature sensors type LM35. All the electronic devices are powered from an uninterruptible power supply.

To collect and record data a software system was implemented. It records and displays the sensors temperature, and generates an alarm if they exceed allowed values in places covered.

III. TEMPERATURE MONITORING

This device allows monitoring the temperature in circuit with Freon of air conditioning systems and the environment temperature of Datacenter. Each independent unit is provided with an element of control by user request or inhibits (canceled) triggering an ALARM possible situation. The program will provide to be possible an alarm only if the drive is powered, knowing that at one time and under favorable conditions, even a single unit in function can ensure optimal thermal regime in the area served. If the request for alarm is activated, exceeding prescribed threshold values of temperature and fan failure, the alarm will be notification. This event will be marked by lighting the lamp unit serves compromised signaling, marking the animation GUI lamp ignition and / or by turning an acoustic alarm device. Alarm triggered can be canceled only by using of the program with a computer mouse click on the indicator light activated representation. This will automatically delete the request of an alarm indicator when canceling because, due to the onset of failure is eliminated and the system still not working within normal parameters. In case of an alarm trigger conditions are not fulfilled yet, but for various reasons wish to launch such an event, such as testing or notification of further events unmonitored, it can trigger the process described in the program before execution Click the mouse pointer on the image desired idle.

Fig. 2 presents the refrigerant temperature at the inlet and outlet of the cooling unit, in two different days.

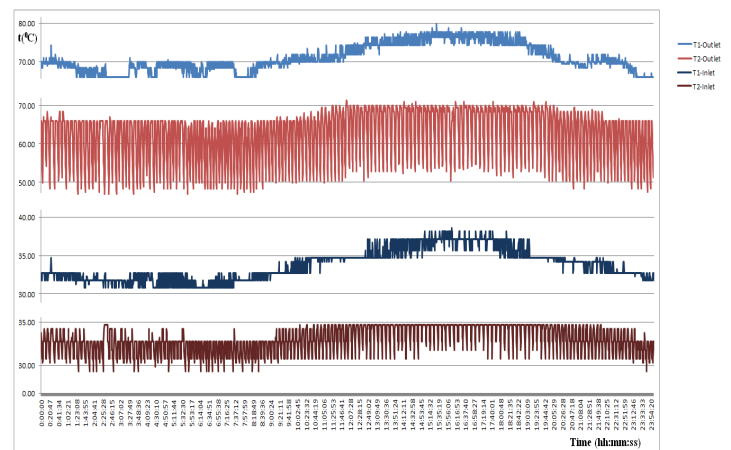


Fig. 2 Refrigerant temperature at the inlet and outlet of the cooling unit, in two different days

It can be observed that during normal working hours, the temperature increase with 5°C approximately (T1 and T2). The ambient temperature was measured in several weeks of summer, at two different air conditioning units and the

variation was not significantly. This fact proves that the cooling air systems work in good conditions.

The ambient temperature in Datacenter was measured with the same device. In this aim, four sensors were installed at four different distances from the floor, in different points of Datacenter, as follows: $h_1 = 2.20$ m, $h_2 = 1.80$ m, $h_3 = 1.50$ m, $h_4 = 0.50$ m. The graphics show that the cooling system is working properly, regardless of the load datacenter, especially during working hours when there is an increase in temperature in the cooling system (Fig. 3).



Fig. 3 Ambient temperature variation in Datacenter

After monitoring the ambient temperature and the cooling system, it was found that the variation is closely related to power consumption. To reduce energy consumption were determined intervals in which some of the computing systems but also some cooling units can be stopped, as needed, without compromising proper functioning of the data center. In fig. 4 are determined time intervals, during which it can intervene to reduce consumption.

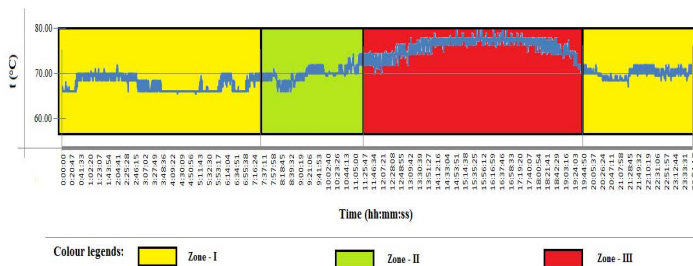


Fig. 4 Test periods for reducing energy consumption

IV. CONCLUSIONS

Following temperature variations that occur both in the cooling system, as well as the ambient Datacenter, can reduce energy consumption. For this purpose, stop some air conditioning units or some computers in different time slots. These intervals were set so that the economy is high, but the work's Datacenter is not disturbed.

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